Experiment No. :- 7

THEORY

- Day Remaining :- It Calculates the number of days left from the Current date until the end of the year.

- Highest_Salary :- It find the maximum Salary
- Lowest_Salary :- It find the minimum Salary.

- Comm > 25% :- It displays employees whose Commission Value is greater than 25% of their Salary.

- Salary in Dollar Format :- It Converts and displays Salary Values in dollar Currency formate using formatting functions.

- Matrix Query :- It display job-wise Salary based on department numbers in matrix from and also Shows the total Salary for each job.

- Last Sunday of Any Month :- It finds the date of the last Sunday for a given month.

- Employees Count (*) :- It display each department number and the total number of employee working in that department.

- Total Sal per dept :- It display each department number and Calculate the total Salary of employees in that department.

```
MariaDB [vijendra0862]> SELECT DATEDIFF(
    ->         STR_TO_DATE(CONCAT(YEAR(CURDATE()),'-12-31'),'%Y-%m-%d'),
    ->         CURDATE()
    ->         ) AS days_remaining;
+----------------+
| days_remaining |
+----------------+
|            308 |
+----------------+
1 row in set (0.001 sec)
```

```
MariaDB [vijendra0862]> SELECT
    -> MAX(sal) AS highest_salary,
    -> MIN(sal) AS lowest_salary,
    -> MAX(sal) - MIN(sal) AS difference
    -> FROM employee;
+----------------+----------------+----------------+
| highest_salary | lowest_salary  | difference     |
+----------------+----------------+----------------+
|           5500 |            880 |           4620 |
+----------------+----------------+----------------+
1 row in set (0.086 sec)
```

```
MariaDB [vijendra0862]> SELECT *
    -> FROM employee
    -> WHERE comm > sal * 0.25;
+-------+-------+----------+------+------------+------+------+--------+
| empno | ename | job      | mgr  | hiredate   | sal  | comm | deptno |
+-------+-------+----------+------+------------+------+------+--------+
|  7654 | MARTIN | SALESMAN | 7698 | 1981-09-28 | 1250 | 1400 |     30 |
+-------+-------+----------+------+------------+------+------+--------+
1 row in set (0.001 sec)
```

```
MariaDB [vijendra0862]> SELECT ename,
    -> CONCAT('$', FORMAT(sal,2)) AS salary_dollar
    -> FROM employee;
+-------+----------------+
| ename | salary_dollar  |
+-------+----------------+
| SMITH | $800.00        |
| ALLEN | $1,600.00      |
| WARD  | $1,250.00      |
| JONES | $2,975.00      |
| MARTIN | $1,250.00      |
| BLAKE | $2,850.00      |
| CLARK | $2,450.00      |
| SCOTT | $3,000.00      |
| KING  | $5,000.00      |
| TURNER | $1,500.00      |
| ADAMS | $1,100.00      |
| JAMES | $950.00        |
| FORD  | $3,000.00      |
| MILLER | $1,300.00      |
+-------+----------------+
14 rows in set (0.001 sec)
```

7 1. Compute the number of days remaining in this year.

```
MariaDB [Vijendra0862]> SELECT DATEDIFF(
                        STR_TO_DATE(Concat(year(Curdate()),'12-31'),
                        '%Y-%m-%d'),
                        CURDATE()
                        ) AS days_remaining;
```

2. Find the highest and lowest salaries and the difference between of them.

```
MariaDB[Vijendra0862]> SELECT
                       Max(sal) As highest_Salary,
                       Min(sal) As lowest_Salary,
                       Max(sal) - Min(sal) AS difference
                       FROM employee;
```

3. List employee whose Commission is greater than 25% of their Salary.

```
MariaDB[Vijendra0862]> Select * FROM employee
                       WHERE Comm > Sal * 0.25;
```

4. Make a query that display Salary in dollar form

```
MariaDB[Vijendra0862]> SELECT ename,
                       CONCAT('$', FORMAT(sal,2)) AS S
                       FROM employee;
```

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```
MariaDB [vijendra0862]> SELECT job,
    -> SUM(CASE WHEN deptno = 10 THEN sal ELSE 0 END) AS dept10,
    -> SUM(CASE WHEN deptno = 20 THEN sal ELSE 0 END) AS dept20,
    -> SUM(CASE WHEN deptno = 30 THEN sal ELSE 0 END) AS dept30,
    -> SUM(CASE WHEN deptno = 40 THEN sal ELSE 0 END) AS dept40,
    -> SUM(sal) AS total_salary
    -> FROM employee
    -> GROUP BY job;
```

job	dept10	dept20	dept30	dept40	total_salary
ANALYST	0	3300	0	3300	6600
CLERK	1430	2090	1045	0	4565
MANAGER	0	5968	3135	0	9103
PRESIDENT	0	5500	0	0	5500
SALESMAN	0	0	5600	0	5600

5 rows in set (0.001 sec)

```
MariaDB [vijendra0862]> SELECT
    -> COUNT(*) AS total_employees,
    -> SUM(YEAR(hiredate)=1980) AS hired_1980,
    -> SUM(YEAR(hiredate)=1981) AS hired_1981,
    -> SUM(YEAR(hiredate)=1982) AS hired_1982,
    -> SUM(YEAR(hiredate)=1983) AS hired_1983
    -> FROM employee;
```

total_employees	hired_1980	hired_1981	hired_1982	hired_1983
14	1	10	2	1

row in set (0.001 sec)

5. Create a Matrix query to display the job, the Salary for that job based on department number and the total Salary for that job for all departments, giving each Column an appropriate heading.

```
MariaDB[Vijendra0862]> SELECT job,
    SUM(Case WHEN deptno=10 THEN Sal ELSE 0 END) AS dept10,
    SUM(Case when deptno=20 then Sal Else 0 END) As dept20,
    SUM(Case when deptno=30 then Sal else 0 END) As dept30,
    SUM(Case when deptno=40 then Sal else 0 end) As dept40,
    SUM(Sal) AS total_Salary
    FROM employee
    GROUP By job;
```

6. Query that will display the total number of employees and of that total the number who were hired in 1980, 1981, 1982 and 1983. Give appropriate Column heading.

```
MariaDB[Vijendra0862]> SELECT
    Count(*) AS total_employees,
    SUM(year(hiredate) = 1980) AS hired_1980,
    SUM(year(hiredate) = 1981) AS hired_1981,
    SUM(year(hiredate) = 1982) AS hired_1982,
    SUM(year(hiredate) = 1983) AS hired_1983.
    FROM employee;
```

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```
MariaDB [vijendra0862]> SELECT DATE_SUB(
    ->         LAST_DAY(CURDATE()),
    ->         INTERVAL (DAYOFWEEK(LAST_DAY(CURDATE()))-1) DAY
    ->         ) AS last_sunday;
+-------------+
| last_sunday |
+-------------+
| 2026-02-22  |
+-------------+
1 row in set (0.001 sec)
```

```
MariaDB [vijendra0862]> SELECT deptno, SUM(sal) AS total_salary
    -> FROM employee
    -> GROUP BY deptno;
+--------+--------------+
| deptno | total_salary |
+--------+--------------+
|     10 |         1430 |
|     20 |        16858 |
|     30 |         9780 |
|     40 |         3300 |
+--------+--------------+
4 rows in set (0.001 sec)
```

```
MariaDB [vijendra0862]> SELECT job, COUNT(*) AS total_employees
    -> FROM employee
    -> GROUP BY job;
+-----------+-----------------+
| job       | total_employees |
+-----------+-----------------+
| ANALYST   |               2 |
| CLERK     |               4 |
| MANAGER   |               3 |
| PRESIDENT |               1 |
| SALESMAN  |               4 |
+-----------+-----------------+
5 rows in set (0.001 sec)
```

```
MariaDB [vijendra0862]> SELECT deptno, COUNT(*) AS total_employees
    -> FROM employee
    -> GROUP BY deptno;
+--------+-----------------+
| deptno | total_employees |
+--------+-----------------+
|     10 |               1 |
|     20 |               6 |
|     30 |               6 |
|     40 |               1 |
+--------+-----------------+
4 rows in set (0.001 sec)
```

7. Query to get the last Sunday of Any Month

```
MariaDB[Vijendra0862]> SELECT DATE_SUB(
                        LAST_DAY(CURDATE()),
                      INTERVAL (DAYOFWEEK(LAST_DAY(CURDATE()))
                       DAY) AS last_Sunday;
```

8. Display department number and total number of employees working in each department

```
MariaDB[Vijendra0862]> SELECT deptno, COUNT(*) AS total_employees
                       FROM employee
                       GROUP By deptno;
```

9. Display the various jobs and total number of employees within each job group.

```
MariaDB[Vijendra0862]> SELECT job, Count(*) AS total_employees
                       FROM employee
                       GROUP By job;
```

10. Display the depat numbers and total Salary for each department.

```
MariaDB[Vijendra0862]> SELECT deptno, SUM(Sal) AS total_Salary
                       FROM employee
                       GROUP By deptno;
```

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Vijendra 27/02/26